REV	DESCRIPTION	DFTR	CHKR	ENGR APPV	DATE
C1	ADDED SHEETS 2 AND 3	DH	MSG	JH ARRUS	11-4

BLDG 514

SWITCH TANK

SH 2 BLDG 514 -90KV HVPS EI-730-790-00 (SH 2) A

SPR0xxxx

SPR0xxxx

-90KV GND INPUT

-90KV OUTPUT GND

INPUT -90KV GND

SH 3 BLDG 514 -90KV HVPS EI-730-790-00 (SH 3) A

SPR0xxxx

SPR0xxxx

FLOW SWITCH

SPR02926 SPR02925

BLDG 132

GND -90KV

TERMINATION TANK SD-730-790-05

J2 SHIELD RED/BLK ORN GRN/BLK GRN BLU BLK WHT/BLK WHT RED

MANUAL GND SW

SW1

-90KV

GND

KLYSTRON

WAVEGUIDE

PS SHIELD RED WHT BLK WHT/BLK GRN BLU RED/BLK GRN/BLK ORN

LEV-3 OIL-LEVEL

KLYSTRON ARC

BNC-1

SLOT 9 15 2 1 14

RACK B132-13 INPUT DC MODULE

SPR05664

RACK B132-14 X-CONNECT PANELS

30-15 30-2 30-1 30-14

13-12 13-11 13-6 13-7

SPR02924 SPR02923

RACK B132-12 RF LOCAL DISPLAY PANEL

INTERNAL CKT BD

SCR ENABLE SYS STATUS KLY CROWBAR

FIBER OPTIC CABLES

FLASHING LIGHT

W6 PRESS

PL5 DCM MOD

AIM MODULE

DB9

9 CONDUCTOR

SPR02946

SPR02947

SPR02944

SPR02945

BLDG 118

JUNCTION BOX

J4 W6 PRESS | FLASHING LIGHT | PLC DC COM 1 2 3 4 5 6 7 8 9 10

J3 1 2 3 4 5 6 7 8 9 10

FIBER OPTICS

J5 HVPS I | HVPS V | LEV-3 1 2 3 4 5 6 7 8 9 10

BNC-1

J1 1 2 3 4 5 6 7 8 9 10

J2 1 2 3 4 5 6 7 8 9 10

N/C

J6 1 2 3 4 5 6 7 8 9 10

HOFFMAN BOX

TOUCH PANEL DISPLAY TS-1 SPEAR 1 — 4 7 9 10

PLC DC COM SPEAR 1

DB9 1 2 3 4 5 6 7 8 9

KLYSTRON PS MONITOR BD SD-730-793-12 SPEAR 1

TOUCH PANEL DISPLAY TS-1 SPEAR 2 — 4 7 9 10

PLC DC COM SPEAR 2

DB9 1 2 3 4 5 6 7 8 9

KLYSTRON PS MONITOR BD SD-730-793-12 SPEAR 1

B (SH 2) RIGHT SIDE INTRFC BD (J2) AND LEFT SIDE INTRFC BD (J2 & J3) SPEAR 1

B (SH 3) RIGHT SIDE INTRFC BD (J2) AND LEFT SIDE INTRFC BD (J2 & J3) SPEAR 2

SPR02924 SPR02923 POWER SUPPLY MONITORS TS-3 SPEAR 1

D (SH 2)

7 8 TS-6 SPEAR 1

E (SH 2)

7 8 TS-6 SPEAR 2

E (SH 2)

SPR02924 SPR02923 POWER SUPPLY MONITORS TS-3 SPEAR 2

D (SH 3)

LEFT SIDE INTERFACE BD BNC-1 SPEAR 1

C (SH 2)

LEFT SIDE INTERFACE BD BNC-1 SPEAR 2

C (SH 3)

GND TANK TS-6 [9-14] SPEAR 1

E (SH 2)

GND TANK TS-6 [9-14] SPEAR2

E (SH 3)

GND TANK TS-6 [1-6] SPEAR 1

E (SH 2)

GND TANK TS-6 [1-6] SPEAR 2

E (SH 3)

13 14 TS-4 SPEAR 1

13 14 TS-4 SPEAR 2

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ENGR J. ARRUDA	10/7/03	J. DIPIRRO
DFTR W. GRECKI	10/14/03	
CHKR J. CZEWSKI	10/7/03	J. OLSEN

CAD FILE NAME: EI444636000101.DGN | SHEET 1 OF 3

SPEAR 3 2.5 MW KLYSTRON POWER SUPPLY SYSTEM ELECTRICAL INTERCONNECT

EI-444-636-00-C1 | E

SPEAR3 2.5MW KLY PS SYS EI

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REV	DESCRIPTION	DFTR	CHKR	ENGR APPV	DATE

SSRL SUBSTATION B507
ID-439-720-20
BUS2, 2000A, 12.47KV
CB160

PANEL: DC 507
VOLT: 120 VDC
CB: 5

PANEL: HVPS 2
VOLT: 208 VAC
CB: 2,4,6

PANEL: HVPS 2
VOLT: 208 VAC
CB: 8,10

BLDG 514

RACEWAY
12.47 KV OVERHEAD SPLICE

PLUG OUTLET

FAN CONTROL ENCLOSURE
8 FANS

BLDG 221
TEMP DETECTOR
AUDIBLE ALARM

KIRK KEY
DISCONNECT SWITCH
FUSES 50E
SURGE ARRESTORS
TO HVPS SEE EI-730-790-00
VACUUM CONTACTOR

VACUUM CONTACTOR DISCONNECT
ID-308-801-06
GP-439-704-02

VACUUM CONTACTOR LOCK-OUT-RELAY

HVPS
EI-730-790-00

TB1

SLAC-1
SHIELD 1
BLK 2
WHT 3
BLK 4
WHT 5
ORG 6
WHT/BLK 7
GRN/BLK 8
ORG/BLK 9
RED 10
RED/BLK 11
RED/WHT 12
BLU 13
BLU/BLK 14
BLU/WHT 15
BLK/WHT 16
GRN/WHT 17
GRN 18

NWL TRANSFORMER

SCR TRIGGERS
HOFFMAN 12"X10"
MS3108E24-20S
MS3108E24-20S
TOP OIL A
BOTTOM OIL B
THERMOCOUPLES
LEV-1 OIL-LEVEL

CROWBAR
HOFFMAN 12"X10"
XFMR ARC P6
DC VOLTAGE 2 P8
DC VOLTAGE 1 P7
LEV-2 OIL-LEVEL
MS3108E18-1S P5
SPR02930 COAX RG-58

GND
-90KV

SH 1 BLDG 514 SWITCH TANK INPUT A (SH 1)
SH 1 JUNCTION BOX BLDG 118 FIBER OPTIC B (SH 1)
SH 1 JUNCTION BOX BLDG 118 BNC-1 C (SH 1)
SH 1 JUNCTION BOX BLDG 118 J5 D (SH 1)
SH 1 JUNCTION BOX BLDG 118 J1, J5 LEV-3, J2 E (SH 1)

SPR0xxxx
SPR02947
SPR02924
SPR02923
SPR02946
SPR02945
SPR02944

TB3
VACUUM CONTACTOR DISCONNECT
TEMP
PPS MONITOR
CONTACTOR READY
CONTACTOR CLOSED
OVERCURRENT
BLOCKING RELAY
PPS
ENABLE
RR
K-4
ON/OFF MX
NEUTRAL-M116,N,Y
ØA:120VAC,M117
ØB:120VAC, X1
ØC:120VAC, X

TS-NWL
OIL PUMP
PRESSURE RELIEF SWITCH
OIL OVERTEMP SWITCH
SPR05701
TRANSFORMER INTERLOCKS
NWL #39308
NWL- TRANSFORMER

DRY AIR
LCW INLET
WATER RETURN
iBLDG 122
TRESTLE
PRESSURE SWITCH NO C
VENTURI
FLOW SWITCH NO NC C
SPR02935 BELDEN 83715 15C *16 TEFLON

SPR02934 BELDEN 83715 15C 15C *16 TEFLON
SPR02933 BELDEN 83715 15C *16 TEFLON
SPR02939
SPR02940
SPR02938 BELDEN 83715 15C *16 TEFLON
SPR02931 BELDEN 83715 15C *16 TEFLON
SPR02929
SPR02941
SPR02931 BELDEN 83715 15C *16 TEFLON
SPR02932

FIBER OPTIC CABLES

T20-1-509 THERMOCOUPLE
BELDEN 88761 SHLD TW PR
XFMR ARC BNC-2
SCR ENABLE J2
SYS STATUS J2
KLY CROWBAR J3
KLYSTRON ARC BNC-1
RIGHT SIDE INTERFACE BOARD
LEFT SIDE INTERFACE BOARD

SPR02927
SPR02928
TS-3
PWR SPLY MONITORS

PHASE SCR'S
PHASE SCR'S
LF SCR CROWBAR DRIVER

TS-5
CONTACTOR CONTROLS
PPS MONITOR
CONTACTOR READY N.C.
CONTACTOR READY N.O.
CONTACTOR CLOSED
INTERLOCK COMMON
BLOCKING RELAY
OVERCURRENT
ENABLE
PPS COM
COMMAND ON/OFF
COMMAND COMMON

SPR05700
N/C
TS-4
OIL LEVEL LOW
SUDDEN PRESSURE
TEMPERATURE
AC CURRENT
TRANSFORMER MONITORS
TS-7

TS-8
PERMITS

COM 10+ IN SLOT 7
AB-1794-IV16

TS-1
PPS GOB1288PNE J1 P1
SPR04549
P2 J2 INTERFACE PANEL J1 P1
RACK 118-46

GND TANK
TS-6
AIR TEMP TC-4
AB-1794-THERMC
SLOT3

HOFFMAN BOX 34" X 42"
WD-730-790-01
WD-730-790-02

SPR02942
PWR SUPPLY CONTROLS
TOUCH PANEL DISPLAY
120VAC
H N GND 120VAC PLUG OUTLET
CKT/20/34 PNL LPB
CKT/34 PNL LPB

SPR04548
OPERATOR LOCK BOX
KIRK KEY
CB20
N GND N H 120 VAC
PANEL BOARD LPB
B117 CLOSET
TB8
PERSONEL PROTECTION SYSTEM
RACK 117-38
BLDG 117

BLDG 118

SPEAR 1

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		DATE	APPROVAL
ENGR	J. [illegible]	12/7/03	J. DIPIRRO
DFTR	W. GORECKI	12/4/03	
CHKR	J. [illegible]TEWSKI	12/7/03	J. OLSZEWSKI

CAD FILE NAME: EI444636000102.DGN | SHEET 2 OF 3

SPEAR 3 2.5 MW KLYSTRON POWER SUPPLY SYSTEM ELECTRICAL INTERCONNECT

EI-444-636-00-C1 | E

REV	DESCRIPTION	DFTR	CHKR	ENGR APPV	DATE

SSRL SUBSTATION B507
ID-439-720-20
BUS2, 2000A, 12.47KV
CB160

PANEL: DC 507
VOLT: 120 VDC
CB: 5

PANEL: HVPS 2
VOLT: 208 VAC
CB: 1,3,5
N ØA ØB ØC

PANEL: HVPS 2
VOLT: 208 VAC
CB: 7,9
Ø1 208 VAC

FAN CONTROL ENCLOSURE
8 FANS

BLDG 221
TEMP DETECTOR
AUDIBLE ALARM

BLDG 514

RACEWAY
12.47 KV OVERHEAD SPLICE
PLUG OUTLET

KIRK KEY
DISCONNECT SWITCH
FUSES 50E
SURGE ARRESTORS
TO HVPS SEE EI-730-790-00
VACUUM CONTACTOR

VACUUM CONTACTOR DISCONNECT
ID-308-801-06
GP-439-704-02

VACUUM CONTACTOR LOCK-OUT-RELAY
TBD

HVPS
EI-730-790-00
TB1

VACUUM CONTACTOR DISCONNECT
TB3
TEMP
PPS MONITOR
CONTACTOR READY
CONTACTOR CLOSED
OVERCURRENT
BLOCKING RELAY
PPS
ENABLE
RR
K-4
ON/OFF
MX
GRN RED
BLU GRN/BLK
GRN/BLK ORG
BLK/WHT GRN/BLK
BLU/BLK GRN/BLK
RED/WHT BLK
RED/BLK GRN/WHT
WHT
NEUTRAL-M116,N,Y
ØA:120VAC,M17
ØB:120VAC, X1
ØC:120VAC, X

TRANSFORMER INTERLOCKS
NWL #39308
NWL- TRANSFORMER
TS-NWL
RED GRN BLU
OIL PUMP
PRESSURE RELIEF SWITCH
OIL OVERTEMP SWITCH
ORG/BLK ORG WHT/BLK BLK/WHT GRN/BLK GRN/WHT BLU/BLK BLU/WHT BLU/WHT RED/BLK RED/WHT
SPR05701

SLAC-1
SHIELD BLK WHT BLK WHT ORG WHT/BLK GRN/BLK ORG/BLK RED RED/BLK RED/WHT BLU BLU/BLK BLU/WHT BLK/WHT GRN/WHT GRN
NWL TRANSFORMER

SCR TRIGGERS
HOFFMAN 12"X10"
TOP OIL A
BOTTOM OIL B
TC-1
TC-2
T20-1-509
THERMOCOUPLES
SHIELD GRN BLU BLU/WHT BLK/WHT RED RED/WHT RED/BLK GRN/WHT GRN/BLK
P1 MS3108E24-20S
P2 MS3108E24-20S
LEV-1 OIL-LEVEL
BLK 2 RED 1

CROWBAR
HOFFMAN 12"X10"
LEV-2 OIL-LEVEL
BLK 2 RED 1
XFMR ARC P6
DC VOLTAGE 2 P8
DC VOLTAGE 1 P7
P5 MS3108E18-1S
SPR02930 COAX RG-58

GND
-90KV
SPR0xxxx
SPR0xxxx

SH 1 BLDG 514 SWITCH TANK INPUT A (SH 1)
SH 1 JUNCTION BOX BLDG 118 FIBER OPTIC B (SH 1)
SH 1 JUNCTION BOX BLDG 118 BNC-1 C (SH 1)
SH 1 JUNCTION BOX BLDG 118 J5 D (SH 1)
SH 1 JUNCTION BOX BLDG 118 J1, J5 LEV-3, J2 E (SH 1)
SPR02947
SPR02924
SPR02923
SPR02946
SPR02945
SPR02944

DRY AIR
LCW INLET
WATER RETURN
iBLDG 122
TRESTLE
PRESSURE SWITCH NO C
VENTURI
FLOW SWITCH NO NC C
SPR02935 BELDEN 83715 15C *16 TEFLON

SPR02934 BELDEN 83715 15C 15C *16 TEFLON
SPR02933 BELDEN 83715 15C *16 TEFLON
SPR02939 SPR02940
SPR02938 BELDEN 83715 15C *16 TEFLON
SPR02931 BELDEN 83715 15C *16 TEFLON
SPR02929 SPR02941
SPR02931 BELDEN 83715 15C *16 TEFLON
SPR02932
FIBER OPTIC CABLES

SPR05700
RED GRN BLU ORG/BLK ORG WHT/BLK BLK/WHT GRN/BLK GRN/WHT BLU/BLK BLU/WHT RED/BLK RED/WHT WHT BLK SHIELD
N/C
TS-4
OIL LEVEL LOW
SUDDEN PRESSURE
TEMPERATURE
AC CURRENT
TRANSFORMER MONITORS
TS-7
GRN BLK/WHT GRN/WHT BLU/WHT BLU/BLK BLU RED/WHT RED/BLK RED ORG/BLK GRN/BLK WHT/BLK ORG WHT BLK SHIELD

PHASE SCR'S
PHASE SCR'S
LF SCR CROWBAR DRIVER
WHT/BLK BLU/WHT BLK/WHT BLK RED RED/WHT RED/BLK GRN/WHT

T20-1-509 THERMOCOUPLE
BELDEN 88761 SHLD TW PR
BELDEN 88761 SHLD TW PR
XFMR ARC BNC-2
J2 SCR ENABLE
RIGHT SIDE INTERFACE BOARD
J2 SYS STATUS
J3 KLY CROWBAR
KLYSTRON ARC BNC-1
LEFT SIDE INTERFACE BOARD
SPR02927
SPR02928
TS-3
PWR SPLY MONITORS

PPS MONITOR
CONTACTOR READY N.C.
CONTACTOR READY N.O.
CONTACTOR CLOSED
INTERLOCK COMMON
BLOCKING RELAY
OVERCURRENT
ENABLE
PPS COM
COMMAND ON/OFF
COMMAND COMMON
RED GRN BLU ORG/BLK ORG WHT/BLK BLK/WHT GRN/BLK BLU/BLK BLU/WHT RED/WHT BLK RED/BLK GRN/WHT
TS-5
CONTACTOR CONTROLS

TS-8
WHT/BLK RED/BLK GRN ORG RED BLU BLK WHT
PERMITS

COM 10+ IN SLOT 7
AB-1794-IV16

TS-1
GRN WHT BLK
PPS GOB1288PNE
J1 P1
SPR04549
P2 J2
INTERFACE PANEL
J1 P1
RACK 118-46

GND TANK
TS-6
ORG RED/BLK WHT/BLK BLK GRN BLU RED WHT GRN/BLK BLK WHT/BLK ORG RED/BLK GRN RED
BLK RED BLK RED

AIR TEMP TC-4
AB-1794-THERMC
SLOT3

HOFFMAN BOX 34" X 42"
WD-730-790-01
WD-730-790-02

SPR02942
PWR SUPPLY CONTROLS
TS-8
WHT/BLK RED/BLK GRN ORG RED BLU BLK WHT
TOUCH PANEL DISPLAY
GND N H 120VAC
H N GND 120VAC
PLUG OUTLET
CKT/20/34 PNL LPB
CKT/34 PNL LPB

SPR04548
OPERATOR LOCK BOX
KIRK KEY
CB34
N GND N H 120 VAC
PANEL BOARD LPB
B117 CLOSET
TB8
PERSONEL PROTECTION SYSTEM
RACK 117-38

BLDG 118
BLDG 117

SPEAR 2

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	DATE	APPROVAL
ENGR J. CHANGGOIO	1/03	J. DIPIETRO
DFTR W. GORECKI	8/14/03	
CHKR J. DOLEWSKI	1/03	J. DOLEWSKI

CAD FILE NAME: EI444636000103.DGN | SHEET 3 OF 3

SPEAR 3 2.5 MW KLYSTRON POWER SUPPLY SYSTEM ELECTRICAL INTERCONNECT

EI-444-636-00-C1 | E

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